

Grade 8  
Unit 12  
Lesson 5 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

For questions 1-5, consider the monomial terms  $18ab^2$  and  $9a^2b^2$ .

1. Write all the factors that the coefficients share.
2. Write all the variables that the terms share.
3. Write all the common factors of the monomials.
4. Complete the area model to find an equivalent expression to  $18ab^2 + 9a^2b^2$  in factored form. After completing the area model, write the equivalent expression.

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|  |          |           |
|--|----------|-----------|
|  | $18ab^2$ | $9a^2b^2$ |
|--|----------|-----------|

5. Verify that the expression in question 4 is equivalent to  $18ab^2 + 9a^2b^2$  using the distributive property.

For questions 6-7, complete the area models for  $8df^2 - 36d^2f^2$  to find two equivalent expressions in factored form. After completing the area models, write the equivalent expressions.

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- 7.

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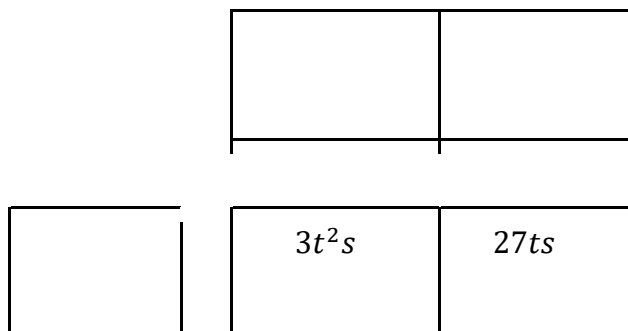
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|  |         |             |
|--|---------|-------------|
|  | $8df^2$ | $-36d^2f^2$ |
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|  |         |             |
|--|---------|-------------|
|  | $8df^2$ | $-36d^2f^2$ |
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For questions 8-9, consider the expression  $3t^2s + 27ts$ .

8. Complete the area model to find an equivalent expression to  $3t^2s + 27ts$  in factored form. After completing the area model, write the equivalent expression.



9. Verify that the expression in question 8 is equivalent to  $3t^2s + 27ts$  using the distributive property.

10. Complete the area model to find an equivalent expression to  $-4r^2s + 20rs^2$  in factored form. After completing the area model, write the equivalent expression.

